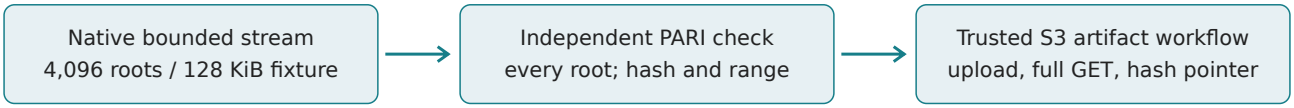
S3-backed native root-chunk fixture

1 November 2026. The native producer writes a bounded stream of baby-root coordinates for the frozen 80-bit kernel. A 4,096-root fixture occupies 131,072 bytes; PARI independently checked every coordinate. This is a development/storage check, not an isogeny performance experiment.

Bounded root preparation and immutable S3 storage



Storage workflow only; no full polynomial, quotient, point map or ECDLP run

Exact data and checks

The source and kernel are the unchanged values in `../parameters.json`. The fixture covers ordinals 0 through 4,095, with kernel indices 2 × ordinal + 1. Each x-coordinate occupies 32 unsigned big-endian bytes. `../s3-chunks/chunk-manifest.json` binds the field, kernel, interval, encoding, producer source and blob digest. `independent-check.json` records the separate PARI arithmetic check for all 4,096 roots.

Local checks passed: byte-identical regeneration, offset-resume agreement with the original tail, rejection of the count/window bounds, and exclusive output creation preventing accidental overwrite. No wall-time advantage or peak-RSS measurement is inferred from these correctness checks.

Storage and completion evidence

The authorized artifact-autofix workflow uploads the 128-KiB object to the existing write-once bucket, reads back the whole object and commits its SHA-256 pointer. The creation-time manifest records upload as pending; only the workflow's pointer commit establishes completed S3 readback. Retrieve with the repository artifact tool and verify the adjacent checksum. No credential values or memory dumps belong in this fixture.

What S3 does and does not establish

Bounded root preparation avoids allocating the entire root vector. It does not make the existing contiguous FLINT polynomials operate on S3. A full external-memory schedule for product/remainder trees, intermediate manifests, resumption and complete point evaluation is still required. S3 is object storage, not process RAM or transparent swap.

The full baby window contains 427,154,957,034 roots. This simple encoding would require 13,668,958,625,088 bytes before intermediates. Only the small fixture has been generated; no full-window job or multi-terabyte upload was started. Future measurements must include construction, transfer, I/O, serialization, failed epochs and verification, following the existing native EKS/checkpoint rules and `../MEASUREMENT_PROTOCOL.md`.

Sources: existing `sqrt_velu.cpp` / `CONSTRUCTOR.md` in the related CM study, the checked `search80.c` point arithmetic, the PARI elliptic-curve API and `docs/artifacts.md` upload/full-byte-readback contract. No new algebraic identity, completed crossing or endpoint measurement is claimed.